

S1 Appendix

Critical slowing down indicators and significance calculation

The restoring rate λ , the variance and the autocorrelation are calculated in the same manner as in Boers 2021. Each time series is first non-linearly detrended using a running mean with a 50-year window. The edges are not removed, so the detrending method is less certain at the first and last 25 years of the time series. The CSD indicators are then calculated in 60-year running windows. The restoring rate of the underlying continuous dynamics is defined by the equation

$$dX_t = \lambda' X_t dt + \sigma dW_t,$$

which results from linearizing the dynamics around a given stable equilibrium, where dW_t denotes the increments of a Wiener process. Note that as in B21 the λ plotted in this study is the numerical result of the regression of the discrete increments $\Delta X_t = X_{t+1} - X_t$ against X_t , and so is related to the analytical, continuous λ' by $\lambda = e^{\lambda'} - 1$ (when the timestep $\Delta t = 1$). As the magnitude of λ is immaterial in this study and we are only concerned with its increase or decrease, both definitions behave similarly and are thus interchangeable for our purposes. In order to test the statistical significance of the linear slopes of CSD indicators, Fourier surrogates are created from the detrended time series. These are calculated by taking the discrete Fourier transform of the time series, multiplying by random phases and then taking the inverse Fourier transform. By the Wiener-Khinchin theorem, the variance and autocorrelation function of wide-sense-stationary random processes are specified by the squared amplitudes of the (discrete) Fourier transform. Thus the Fourier surrogates preserve the variance and autocorrelation function of the original time series. The indicators are then calculated from the surrogates and the trends of these form the distribution that is used to calculate the p value of the model run indicator trend. For the calculations in the main text that use a 60 year window, 1000 Fourier surrogates are generated for each time series. For the checks on the 50 and 70 year window lengths, 100 Fourier time surrogates are generated for each time series for reasons of computational expense.